NAGA VENKATA SAI CHEJARLA

Aspiring VLSI Design & Verification Engineer
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PROFESSIONAL SUMMARY

Electronics and Communication Engineering graduate (B.Tech, 2026) with a strong foundation in digital electronics, CMOS design and Verilog/VHDL. Hands-on experience in RTL coding, functional simulation, testbench development and waveform-based timing verification using ModelSim, QuestaSim and Xilinx tools through a VLSI internship and ongoing design verification training at VLSI Guru Institute. Seeking an entry-level Design Verification-RTL Design role.

EDUCATION

St. Ann's College of Engineering and Technology, Chirala, Andhra Pradesh **2022 - 2026**
Bachelor of Technology in Electronics and Communication Engineering | Percentage: 71%

Sri Vignani VGRM Junior College, Andhra Pradesh **2020 - 2022**
Intermediate (PMC) | Percentage: 81%

TECHNICAL SKILLS

HDLs & Programming: Verilog HDL, VHDL, C, Embedded C, Python (Basics)

EDA & Simulation Tools: ModelSim, QuestaSim, Xilinx Vivado, Xilinx ISE Design Suite, DSCH 3.5, Microwind 3.1, Logisim

Platforms & Editors: Linux, GVim, Arduino IDE

VLSI & Design Concepts: Digital Electronics, CMOS Design, RTL Design, Finite State Machines, Parallel Prefix Adders, Low-Power Design, Testbench Development, Functional Simulation, Waveform Analysis, Timing Verification

Hardware & Embedded: NodeMCU, Arduino, RFID, IR and Vibration Sensors, GPS Module, PCB Design Basics

Soft Skills: Communication, Adaptability, Team Management, Problem Solving

Currently Learning: SystemVerilog, Verification Methodology Fundamentals (UVM)

INTERNSHIP & TRAINING

VLSI Signal Processing Intern | *Shri Shasha Prayathi Technologies Pvt. Ltd.*

- Designed and verified digital circuits using Verilog HDL for VLSI signal processing applications.
- Implemented RTL code and functional simulation models for core DSP arithmetic modules.
- Performed waveform analysis and testbench verification to validate circuit timing.

Design Verification Training | *VLSI Guru Institute* **Ongoing**

- Training in RTL simulation and verification flow using ModelSim, QuestaSim, GVim and Linux.

PROJECTS

Hierarchical Full-Adder-Based 16-bit Ripple Carry Adder Architecture in Logisim | *Logisim*

- Built a 16-bit ripple carry adder by cascading full adders at gate level and verified sum and carry outputs across test input combinations.

Optimization of 4-bit Ladner-Fischer Parallel Prefix Adder for Low-Power VLSI Applications | *DSCH 3.5, Microwind*

- Designed a 4-bit Ladner-Fischer parallel prefix adder at schematic level and optimized the prefix network for reduced logic depth and power.
- Created the CMOS layout and analyzed timing and power behavior to validate suitability for low-power applications.

FSM-Based Traffic Light Controller: Dual-HDL Implementation in Verilog and VHDL | *Verilog HDL, VHDL, Vivado*

- Modeled the controller as a Finite State Machine in both Verilog and VHDL with timer-based state transitions.
- Simulated the design in Vivado and verified state sequencing using testbench waveforms.

Design and Functional Verification of an 8-bit ALU using Verilog HDL | *Verilog HDL, Xilinx ISE Design Suite*

- Designed an 8-bit ALU supporting arithmetic and logical operations and verified each operation with a self-written testbench and waveform analysis.

Synchronous FIFO Memory Design and Verification | *Verilog HDL, ModelSim, Linux, GVim* **Ongoing**

- Developing a parameterized synchronous FIFO with full/empty flag logic and verifying it with a testbench in ModelSim.

UART Transmitter and Receiver using FSM | *Verilog HDL, ModelSim/QuestaSim, Linux, GVim* **Ongoing**

- Developing FSM-based UART transmitter & receiver modules and verifying serial data transfer through waveform analysis.

CERTIFICATIONS

- VLSI Design Workshop (3-Day) - KALKI TECH | Spoken Tutorial Certification - C, Python | PCB Designing

ACTIVITIES and ACHIEVEMENTS

- NSS Commando (2022 - 2026): Active volunteer in National Service Scheme community service activities.
- Actively participated in project expos and fest events, including paper presentations.